

Street-View Image Retrieval for Natural Language-Guided User Localization in Urban Environments

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1. Abstract

Autonomous personal mobility and robotic delivery services require reliable user localization for dispatching robots to appropriate pickup or drop-off points. However, GNSS signals can degrade in urban canyons or modality-restricted settings. We propose UrbanLoc, a two-stage language-guided user localization method that retrieves geotagged street-view images and reranks candidates using object coverage. UrbanLoc leverages complex vision-language cues, including storefront signage, through location-aware alignment with area summaries and MLLM-based reranking. We also construct a benchmark with 34k descriptions for 15k locations across 200 km² regions. Experiments show that UrbanLoc outperforms baseline methods.

2. Introduction

Language-guided user localization offers a hardware-agnostic alternative when conventional sensing modalities are unavailable: GNSS signals degrade in urban canyons, and WiFi fingerprinting or cameras [10], [21] may be restricted by connectivity limitations or privacy concerns [14]. In this study, we focus on predicting a user's target location from a natural language description of their surroundings [6], [15], [16], [17]. Fig. 1 illustrates a typical scenario: the user requests a pickup by describing the environment (e.g., "I am standing under the traffic light at the intersection lined with tall brown buildings, right in front of the white shop with the 'LOS TACOS' sign"), and UrbanLoc estimates the location so that a robot can be dispatched.

This task is challenging because users typically refer only to salient landmarks or sign text, creating a many-to-many correspondence between descriptions and candidate locations. Conventional methods rely on point clouds [15], [16] or satellite images [17], which often lack fine-grained vision-language cues (e.g., storefront signage and spatial layout) and do not model area-level features shared across a street.

To address these limitations, we propose UrbanLoc, a two-



Fig. 1 Typical scenario for this task. The user requests a pickup from a location where sensing modalities are restricted by describing the surrounding environment using a smartphone. UrbanLoc predicts the user's location, and a robot is dispatched to the estimated destination.

stage language-guided user localization method via multimodal retrieval and MLLM-based reranking over geotagged street-view images. We introduce location-aware vision-language alignment with area summaries and MLLM-based reranking using object coverage, the latter of which suppresses candidates that are semantically plausible yet geographically unrelated.

The main contributions of this study are as follows:

- We propose UrbanLoc, a two-stage language-guided user localization method via multimodal retrieval over geotagged street-view images.
- We introduce location-aware vision-language alignment using area summaries, along with MLLM-based reranking using object coverage.
- We construct the UrbanLoc-Text benchmark, which consists of 34k descriptions referring to 15k locations across 200 km² regions.

3. Methodology

We focus on the task of predicting a user's target location from a natural language description of their surroundings. We assume that (i) the user is in a densely built urban area where meter-level localization is unavailable, and (ii) a service provider has access to a collection of geotagged street-view images.

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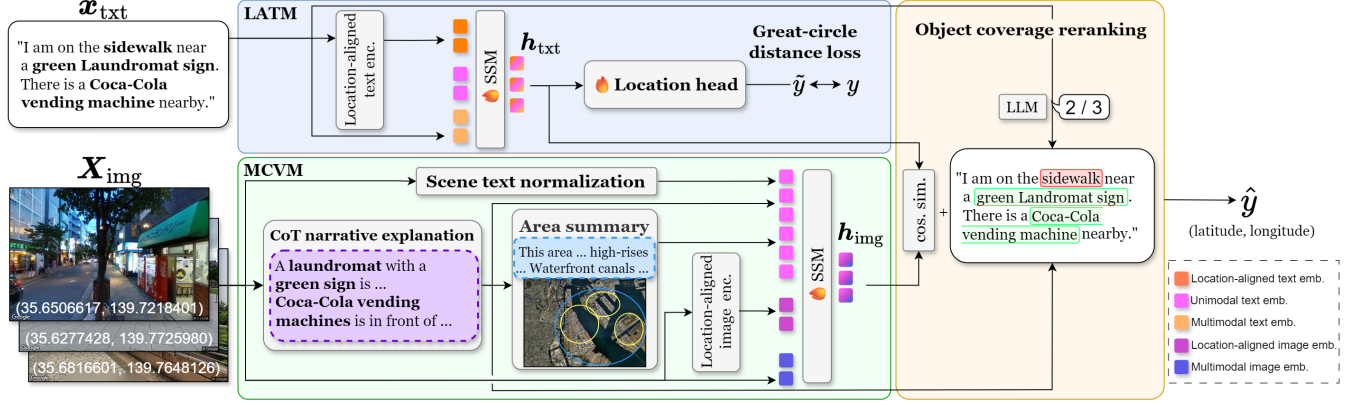


Fig. 2 Architecture of UrbanLoc. LATM obtains location-aware *text* representations by introducing an auxiliary location head. Location-aware *visual* representations are obtained in MCVM by aggregating cues shared across a street. Finally, UrbanLoc reranks candidate locations based on object coverage. $\tilde{y}$, $\hat{y}$, and y denote the location predicted by the auxiliary location head, the final estimated location, and the ground-truth location, respectively. LATM, MCVM, and SSM represent Location-Aware Text Module, Multi-Cue Visual Module, and a deep state space model encoder, respectively.

We propose UrbanLoc, a two-stage language-guided user localization method formulated as text-to-image retrieval over geotagged street-view images, enabling appearance-dependent cues to be grounded against candidate images.

Fig. 2 shows the architecture of UrbanLoc. Our model consists of Location-Aware Text Module (LATM) and Multi-Cue Visual Module (MCVM), together with object coverage reranking. LATM obtains a location-aware text representation using an auxiliary location head with a great-circle distance loss. MCVM includes (i) a chain-of-thought (CoT) narrative representation capturing object-level attributes and (ii) an area-level summary capturing characteristics shared across nearby locations. UrbanLoc reranks candidate locations based on object coverage. The model takes $\mathbf{x}_{\text{txt}}$ and $\mathbf{X}_{\text{img}} = \{\mathbf{x}_{\text{img}}^{(i)}\}_{i=1}^{N_{\text{img}}}$ as input, where $\mathbf{x}_{\text{txt}}$, $\mathbf{x}_{\text{img}}^{(i)} \in \mathbb{R}^{3 \times W \times H}$, and $\mathbf{x}_{\text{txt}}^{(j)}$ denote a description, a geotagged image, and the j -th description when indexing multiple descriptions, respectively.

3.1 Location-Aware Text Module

LATM maps $\mathbf{x}_{\text{txt}}$ to a location-aware text representation $\mathbf{h}_{\text{txt}}$. Since $\mathbf{x}_{\text{txt}}$ can match multiple locations, standard vision-language embeddings [8], [9], [19] may not sufficiently distinguish the target from plausible candidates. We therefore introduce an auxiliary location head that estimates geographic coordinates from $\mathbf{x}_{\text{txt}}$, supervised by a great-circle distance regression loss (see Eq. (6)).

We obtain $\mathbf{h}_{\text{txt}}$ by applying location-aligned, unimodal, and multimodal text encoders to $\mathbf{x}_{\text{txt}}$, concatenating the features, and passing them through a deep SSM [5]. The auxiliary head predicts the geographic coordinates $\tilde{y}$.

3.2 Multi-Cue Visual Module

To capture both visual and location-related cues, MCVM combines visual appearance features with geographic features via the following three submodules.

3.2.1 CoT Narrative Representation

Because visual representations alone can miss object-level

and relational semantics, we utilize a narrative representation [4]. To mitigate hallucinations, an MLLM first generates a structured object list from $\mathbf{x}_{\text{img}}^{(i)}$ to ground the explanation, then produces a CoT narrative explanation (purple block in Fig. 2), which a text encoder embeds as the CoT narrative representation.

3.2.2 Area Summary

Since a single image provides only a partial, viewpoint-dependent observation, it is insufficient to capture area-level cues (e.g., architectural styles) useful for localization. We therefore compute an area mean vector from the CoT narrative representations of nearby locations, select representative locations that are closest to the mean, and generate an LLM-based area summary (blue block in Fig. 2) from their explanations, which a text encoder embeds into the area summary feature.

3.2.3 Scene Text Normalization

To handle language mismatches between $\mathbf{x}_{\text{txt}}$ and multilingual OCR text, a multilingual MLLM [18] normalizes detected scene text by translating non-named-entity tokens and converting named entities into brief explanations (e.g., “El Corte Inglés” $\rightarrow$ “a department store”), and a text encoder embeds the result.

Given $\mathbf{x}_{\text{img}}^{(i)}$, we encode it with a location-aligned encoder [2] and a multimodal encoder [8], concatenate their outputs with the three submodule features, and process the result with a deep SSM to obtain $\mathbf{h}_{\text{img}}^{(i)}$. As $\mathbf{h}_{\text{img}}^{(i)}$ is pre-computed offline, it adds no encoding cost at inference time.

3.3 Object Coverage Reranking

A single cosine similarity score can over-rank partial matches when $\mathbf{x}_{\text{txt}}^{(j)}$ specifies multiple objects, as exemplified in Fig. 2. UrbanLoc therefore computes:

$$s(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)}) = \frac{\mathbf{h}_{\text{txt}}^{(j)} \cdot \mathbf{h}_{\text{img}}^{(i)}}{\|\mathbf{h}_{\text{txt}}^{(j)}\| \|\mathbf{h}_{\text{img}}^{(i)}\|}, \quad (1)$$

and reranks $\mathbf{X}_{\text{img}}$ using:

$$s'(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)}) = s(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)}) + \lambda_{\text{obj}} R_{\text{obj}}^{(i,j)}, \quad (2)$$

Table 1 Quantitative comparison of UrbanLoc and the baseline methods on the New York and Tokyo subsets of the UrbanLoc-Text benchmark. The mean and standard deviation over five trials are reported. The highest score for each metric is in **bold**, and the second-highest score is underlined.

[%] Method	New York			Tokyo		
	L@50↑	R@1↑	R@5↑	L@50↑	R@1↑	R@5↑
CLIP [9]	18.9 ± 0.1	18.3 ± 0.1	31.8 ± 0.1	19.3 ± 0.2	18.8 ± 0.2	34.2 ± 0.1
BEiT-3 [13]	17.5 ± 0.2	16.6 ± 0.2	31.6 ± 0.4	17.8 ± 0.2	17.3 ± 0.3	34.7 ± 0.4
BLIP-2 [7]	17.9 ± 0.0	17.7 ± 0.0	34.2 ± 0.1	19.8 ± 0.0	19.4 ± 0.0	37.6 ± 0.1
SigLIP [19]	21.5 ± 0.2	20.7 ± 0.2	35.3 ± 0.2	18.3 ± 0.2	17.8 ± 0.2	34.5 ± 0.1
Long-CLIP [20]	17.2 ± 0.1	16.6 ± 0.1	30.9 ± 0.1	18.4 ± 0.1	17.8 ± 0.1	34.5 ± 0.1
CrossText2Loc [17]	2.5 ± 0.1	2.2 ± 0.0	5.8 ± 0.0	3.2 ± 0.1	2.9 ± 0.0	4.5 ± 0.1
SigLIP 2 [12]	19.7 ± 0.1	18.9 ± 0.0	34.3 ± 0.0	22.7 ± 0.1	22.1 ± 0.1	40.7 ± 0.1
Meta CLIP 2 [3]	15.6 ± 0.1	15.2 ± 0.1	27.8 ± 0.1	18.3 ± 0.1	17.8 ± 0.1	32.8 ± 0.1
Qwen3-VL-Embedding [8]	17.9 ± 0.6	17.2 ± 0.5	31.2 ± 1.0	19.2 ± 0.5	18.8 ± 0.6	32.9 ± 0.8
Qwen3-VL-Embedding + Qwen3-VL-Reranker	<u>26.8</u> ± 0.3	<u>25.9</u> ± 0.4	<u>36.3</u> ± 0.3	<u>26.6</u> ± 0.3	<u>26.0</u> ± 0.3	<u>36.5</u> ± 0.4
UrbanLoc (ours)	30.2 ± 0.2	28.9 ± 0.2	44.9 ± 0.5	31.8 ± 0.4	31.0 ± 0.3	49.4 ± 0.7

where $R_{\text{obj}}^{(i,j)} \in [0, 1]$ is the fraction of objects in $\mathbf{x}_{\text{txt}}^{(j)}$ covered by the CoT narrative of $\mathbf{x}_{\text{img}}^{(i)}$, computed by an LLM that decomposes both into attribute-augmented object sets and matches them. For efficiency, $R_{\text{obj}}^{(i,j)}$ is computed only for the top- k candidates; $R_{\text{obj}}^{(i,j)} = 0$ otherwise. The model outputs $\hat{\mathbf{y}} = \mathbf{y}^{(i^*)}$, where $i^* = \arg \max_i s'(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)})$ and $\mathbf{y}^{(i)}$ denotes the geographic coordinates of $\mathbf{x}_{\text{img}}^{(i)}$.

We use the following loss:

$$\mathcal{L} = \mathcal{L}_{\text{t2i}} + \mathcal{L}_{\text{i2t}} + \lambda_{\text{dist}} \mathcal{L}_{\text{dist}}, \quad (3)$$

$$\mathcal{L}_{\text{t2i}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s(\mathbf{x}_{\text{txt}}^{(i)}, \mathbf{x}_{\text{img}}^{(i)})/\tau)}{\sum_{j=1}^B \exp(s(\mathbf{x}_{\text{txt}}^{(i)}, \mathbf{x}_{\text{img}}^{(j)})/\tau)}, \quad (4)$$

$$\mathcal{L}_{\text{i2t}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s(\mathbf{x}_{\text{txt}}^{(i)}, \mathbf{x}_{\text{img}}^{(i)})/\tau)}{\sum_{j=1}^B \exp(s(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)})/\tau)}, \quad (5)$$

$$\mathcal{L}_{\text{dist}} = \frac{1}{B} \sum_{i=1}^B d_{\text{gc}}(\tilde{\mathbf{y}}^{(i)}, \mathbf{y}^{(i)}), \quad (6)$$

where d_{gc} and $\tilde{\mathbf{y}}^{(i)}$ denote the great-circle distance (Haversine formula [11]) and the auxiliary head output, respectively. In addition, λ_{dist} , B , and τ denote a weighting hyperparameter, a batch size, and a temperature, respectively.

4. Experiments

4.1 Experimental Setup

Benchmark. We constructed UrbanLoc-Text, which contains 33,928 descriptions referring to 14,954 locations collected via Mapillary. Test-set descriptions were provided by 151 annotators (45.3 descriptions per annotator on average); training/validation descriptions were generated using Qwen3-VL [1]. The benchmark was geographically divided into training, validation, and test sets, containing 24,194, 2,894, and 6,840 samples, respectively.

Implementation Details. We trained UrbanLoc for 50 epochs using AdamW with a learning rate of 1×10^{-5} and batch size 256, setting $\lambda_{\text{obj}} = 0.5$, $\lambda_{\text{dist}} = 0.2$, and $k = 20$. The model has approximately 32.7M trainable parameters

and was trained on a GeForce RTX 4090 in approximately 50 minutes.

Baselines. We used CLIP [9], BEiT-3 [13], BLIP-2 [7], SigLIP [19], Long-CLIP [20], SigLIP 2 [12], Meta CLIP 2 [3], Qwen3-VL-Embedding [8], Qwen3-VL-Embedding + Qwen3-VL-Reranker, and CrossText2Loc [17] (OSM images only, due to terms of use) as baselines. All methods were fine-tuned on UrbanLoc-Text.

4.2 Quantitative Results

Table 1 presents the quantitative comparison between UrbanLoc and the baseline methods on the New York and Tokyo subsets of the UrbanLoc-Text benchmark. We used localization recall@ N ($L@N$, $N = 50$) [17] and recall@ K ($R@K$, $K = 1, 5$) as evaluation metrics, with $L@50$ as the primary metric. The mean and standard deviation over five trials are reported.

As listed in Table 1, UrbanLoc achieved the best $L@50$ scores of 30.2% and 31.8% on the New York and Tokyo subsets, respectively, together with $R@1$ scores of 28.9% and 31.0%. In comparison, the strongest baseline obtained $L@50$ scores of 26.8% and 26.6%, and $R@1$ scores of 25.9% and 26.0%. The performance differences between UrbanLoc and the baseline methods were statistically significant ($p < 0.01$).

4.3 Qualitative Results

Fig. 3 shows a representative qualitative result from the Tokyo subset, comparing UrbanLoc with the baseline method (Qwen3-VL-Embedding + Qwen3-VL-Reranker) [8]. Fig. 3(a) shows $\mathbf{x}_{\text{img}}^{(i)}$, $\mathbf{y}$, and a magnified detail view; Fig. 3(b) shows the top-3 predicted locations for each method, arranged in separate rows.

Here, $\mathbf{x}_{\text{txt}}$ was “There is a parking lot on the left with a yellow sign saying max 3,600 yen, and a red sign for a fire hydrant on the right.” The baseline ranked the target at rank 85 ($d_{\text{gc}} = 11.9$ km), whereas UrbanLoc ranked it at rank 1 by leveraging the “3,600 yen” scene text, with all

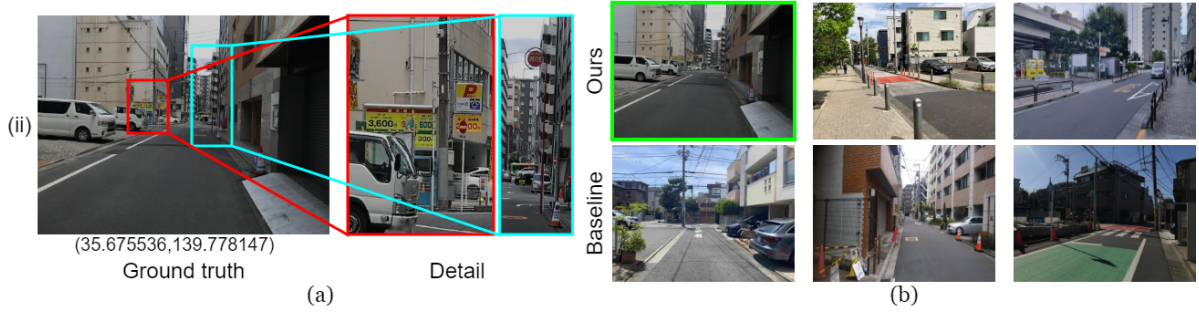


Fig. 3 Qualitative result for a representative successful case from the Tokyo subset. (a) shows $\mathbf{x}_{\text{img}}^{(i)}$, $\mathbf{y}$, and a magnified detail view. (b) shows the top-3 predicted locations for UrbanLoc and the baseline method (Qwen3-VL-Embedding + Qwen3-VL-Reranker), arranged in separate rows.

Table 2 Quantitative results of the ablation studies. The mean and standard deviation over five trials are reported. The best value for each metric is in **bold**.

[%]		New York			Tokyo		
Model		L@50 $\uparrow$	R@1 $\uparrow$	R@5 $\uparrow$	L@50 $\uparrow$	R@1 $\uparrow$	R@5 $\uparrow$
(i)	UrbanLoc (full)	30.2 ± 0.2	28.9 ± 0.2	44.9 ± 0.5	31.8 ± 0.4	31.0 ± 0.3	49.4 ± 0.7
(ii)	w/o location head	28.2 ± 0.3	27.0 ± 0.4	39.8 ± 0.4	28.8 ± 0.1	28.2 ± 0.1	42.0 ± 0.3
(iii)	w/o CoT narrative representation	27.7 ± 0.4	26.4 ± 0.4	39.3 ± 0.7	28.2 ± 0.4	27.6 ± 0.4	40.7 ± 0.3
(iv)	w/o area summary	28.1 ± 0.4	26.9 ± 0.3	40.1 ± 0.2	29.4 ± 0.4	28.8 ± 0.4	42.6 ± 0.4
(v)	w/o scene text normalization	27.6 ± 0.4	26.5 ± 0.3	39.5 ± 0.4	28.9 ± 0.1	28.2 ± 0.1	42.1 ± 0.4
(vi)	w/o object coverage reranking	24.7 ± 0.3	23.6 ± 0.3	40.4 ± 0.5	25.1 ± 0.7	24.4 ± 0.7	43.1 ± 0.7

top-3 predictions containing both a parking lot and its sign. These results indicate that UrbanLoc can identify the target location by leveraging both visual attributes and scene text cues.

4.4 Ablation Study

Table 2 presents the quantitative results of ablation studies. The mean and standard deviation over five trials are reported.

Location head ablation: To investigate the effect of the location head, we removed it from LATM. The L@50 scores of Model (ii) were 2.0 and 3.0 points lower than those of Model (i) on the New York and Tokyo subsets, respectively. These results suggest that incorporating location awareness into text representations through the location head is beneficial for user localization.

CoT narrative representation ablation: To assess the impact of the CoT narrative representation, we removed it to obtain Model (iii). Model (iii) had L@50 scores of 27.7% and 28.2% on the New York and Tokyo subsets, which were 2.5 and 3.6 points lower than those of Model (i), respectively. These results indicate that the CoT narrative representation enriches the visual representation by capturing fine-grained attributes and spatial relationships in each image, thereby improving vision-language alignment.

Area summary ablation: We evaluated the effectiveness of the area summary by omitting it. Table 2 shows that removing this submodule (Model (iv)) decreased L@50 scores by 2.1 and 2.4 points on the New York and Tokyo subsets, respectively, compared with the full model (Model (i)). This indicates that the area summary contributes to

user localization by providing coarse area-level cues.

Scene text normalization ablation: We omitted the scene text normalization submodule to analyze its contribution. Compared with Model (i), Model (v) had L@50 scores that were 2.6 and 2.9 points lower on the New York and Tokyo subsets, respectively. This suggests that scene text normalization enables our model to identify the target location based on multilingual text cues in images.

Object coverage reranking ablation: We assessed the impact of the object coverage reranking by comparing Model (i) with Model (vi), in which $\lambda_{\text{obj}} R_{\text{obj}}^{(i,j)}$ is removed from $s'(\mathbf{x}_{\text{txt}}^{(j)}, \mathbf{x}_{\text{img}}^{(i)})$. Without the object coverage reranking, L@50 scores decreased to 24.7% and 25.1% on the New York and Tokyo subsets, respectively, corresponding to decreases of 5.5 and 6.7 points from the full model. This indicates that object-level agreement is important for improving user localization performance.

5. Conclusions

In this study, we focused on the task of predicting target locations given user-provided descriptions of their surroundings. We proposed UrbanLoc, a two-stage language-guided user localization method via multimodal retrieval over geo-tagged street-view images. We introduced location-aware vision-language alignment using area summaries, along with MLLM-based reranking using object coverage. Furthermore, we constructed the UrbanLoc-Text benchmark, which consists of 34k descriptions referring to 15k locations across 200 km² regions in multiple cities. Experimental results show that UrbanLoc outperformed the baseline methods on standard evaluation metrics.

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